2020B F=ma Exam: Problem 7

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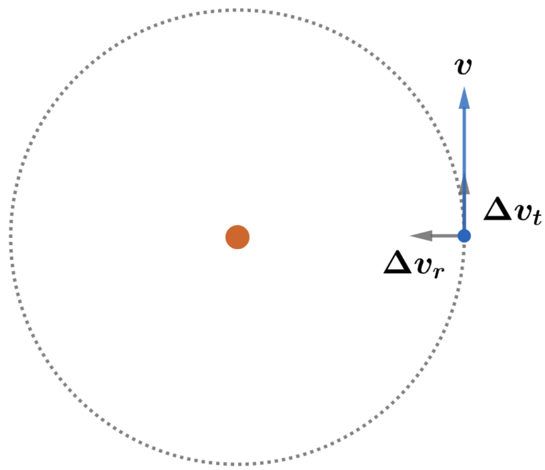

Recall the energy of an elliptical orbit is given by

$$E = -\frac{GMm}{2a}$$

which combined with Kepler's third law $T^2 \propto a^3$ implies both satellites have the same energy since they have the same period. The energy of the first satellite is

$$E_1 = \frac{1}{2}m(v^2 + \Delta v_r^2)$$

The energy of the second satellite is

$$E_2 = \frac{1}{2}m(v + \Delta v_t)^2$$

Since $E_1 = E_2$,

$$v^2 + \Delta v_r^2 = v^2 + 2v\Delta v_t + \Delta v_t^2$$

Since $v \gg \Delta v_t$,

$$\Delta v_r^2 = 2v\Delta v_t$$

$$\Delta v_t = \frac{\Delta v_r^2}{2v} = 0.00005\,\mathrm{m/s}$$

so the answer is $\boxed{\text{A}}$.